

ABHISHEK UPADHYAY

Software Engineer | Python & Backend Developer | AI/LLM Builder

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PROFESSIONAL SUMMARY

Results-driven Software Engineer with hands-on experience building full-stack web applications, AI-powered systems, and automation tools. Proficient in Python, Java, SQL, and modern LLM frameworks. Demonstrated ability to deliver measurable improvements in system performance. Passionate about applying Retrieval-Augmented Generation (RAG), multi-LLM pipelines, and IoT solutions to real-world challenges.

TECHNICAL SKILLS

Languages	Python, Java, SQL, HTML, CSS, JavaScript
AI / LLM	LangChain, RAG, LLaMA, GPT APIs, Prompt Engineering, Vector Embeddings (FAISS/ChromaDB)
Automation	n8n (workflow automation), REST APIs, Webhook integrations
Databases	MySQL, MongoDB, PostgreSQL
Tools	Git, GitHub, VS Code, Postman, Linux
CS Fundamentals	Data Structures & Algorithms, OOP, DBMS, System Design

PROFESSIONAL EXPERIENCE

Python Developer Intern

Mar 2024 – Apr 2024

InternPe Pvt Ltd — Mumbai, IN

- Engineered Python automation scripts that accelerated project processing speed by 20%, reducing manual workload and improving team efficiency across multiple workflows.
- Performed codebase analysis to identify performance bottlenecks; reduced execution time by 15%, enabling faster data processing and better system scalability.
- Presented optimization findings and actionable strategies to senior stakeholders, directly aligning technical improvements with business objectives.

PROJECTS

RAG-Powered Intelligent Q&A; System

2024

Stack: Python · LangChain · FAISS · LLM APIs

- Architected a Retrieval-Augmented Generation (RAG) pipeline enabling users to query large document corpora in natural language, reducing manual search effort by over 60%.
- Integrated vector embeddings (FAISS/ChromaDB) with semantic chunking to retrieve contextually relevant document segments, feeding them into an LLM for source-grounded answers.
- Optimised chunking strategies and retrieval ranking, improving answer relevance by ~35% over naive keyword-based search.

AI Video Maker — Multi-LLM Automation Pipeline

2024

Stack: n8n · LLaMA · GPT APIs · Python · TTS

- Built an end-to-end automated video creation workflow using n8n, orchestrating multiple LLMs (LLaMA local + GPT cloud) to auto-generate scripts, voiceovers, and captions from text prompts.
- Designed modular n8n nodes for script generation, image synthesis prompt creation, and TTS integration, cutting manual video production effort by ~70%.
- Implemented LLM fallback routing — switching between LLaMA (low-latency, free) and GPT (high-quality) based on task complexity, optimising cost and response quality.

LifeStream — Blood Bank Management System

2023

Stack: PHP · MySQL · JavaScript

- Designed a PHP platform with real-time blood inventory tracking, boosting operational efficiency by 30% and improving cross-device data accuracy.
- Implemented role-based secure authentication for donors and hospital staff, reducing unauthorised access incidents by 25%.
- Built an appointment scheduling module that improved donor engagement and appointment adherence by 20%.

HospLink — Hospital Management System

2023

Stack: MySQL · HTML · CSS · OOP Design

- Developed a robust database-driven platform processing 1,000+ patient records daily across multiple departments with 100+ concurrent users.
- Applied OOP design principles to optimise system architecture, reducing data retrieval time by 30%.

SafeGuard IoT — Human Safety Sensor System

2023

Stack: IoT · Embedded Systems · REST API · Push Notifications

- Engineered an IoT sensor system monitoring critical environmental parameters, reducing hazard detection and response times by 40%.
- Developed real-time data transmission protocols between edge sensors and a centralised dashboard, improving reliability by 35%.
- Integrated multi-channel alert system (web + mobile push) for instant safety interventions.

EDUCATION

B.E. — Computer Engineering

Jul 2020 – Jun 2024

Lokmanya Tilak College of Engineering, Mumbai, India

CERTIFICATIONS

- IBM: Python 101 for Data Science
- JP Morgan Chase: Software Engineering Virtual Job Simulation (Forage)

EXTRACURRICULAR & VOLUNTEER

- Organised Tree-Planting Initiative at Adarsh English High School — led team coordination and community outreach.
- Volunteer, College Blood Donation Camp — coordinated logistics and donor management for 100+ participants.